

Unit 5 :: Fundamentals of Data Visualization

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Syllabus

Unit 5: Fundamentals of data visualization

Principles and use cases for categorical and continuous data plots, including column, line, pie, bar, area and scatter plots; customizing chart elements (axes, scales, legends, and labels); fine-tuning aesthetics for clarity and technical accuracy; advanced plotting techniques: 3D surface plots, contour mapping and multi-layered/multipanel figures.

1 Categorical and continuous data plots

A dataset is said to be *categorical* if its independent variable takes one of a small, discrete set of labels (e.g. the day of the week, the floor number of a skyscraper), and *continuous* if its independent variable can take any value within a range (e.g. time, angle, or magnetic field strength). The choice of plot type should follow from which of these two kinds of variable is being plotted, not from visual preference alone.

1.1 Column and bar plots

In a *column plot*, each category is represented as a vertical bar whose height is proportional to a summary value (such as a mean, a total or a count). In a *bar plot*, the idea is the same except that it is drawn using horizontal bars.

Both types of plots are appropriate when the independent variable is categorical. For example, they can be used to compare the mean temperature across different weeks.

1.2 Line plots

1.3 Pie charts

1.4 Area plots

1.5 Scatter plots